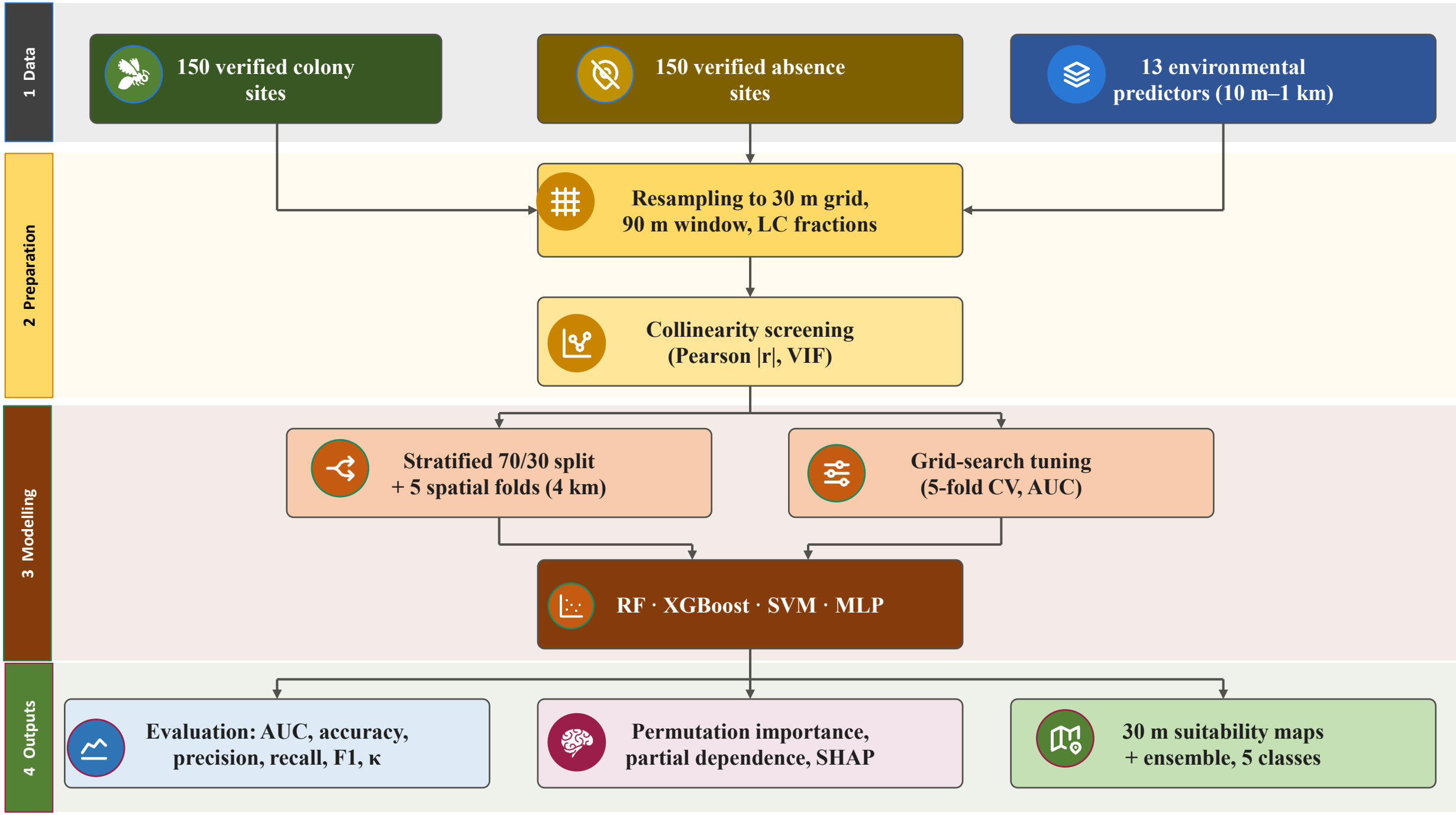

1 Data
150 verified colony sites
150 verified absence sites
13 environmental predictors (10 m–1 km)
2 Preparation
Resampling to 30 m grid, 90 m window, LC fractions
Collinearity screening (Pearson |r|, VIF)
3 Modelling
Stratified 70/30 split + 5 spatial folds (4 km)
Grid-search tuning (5-fold CV, AUC)
RF · XGBoost · SVM · MLP
4 Outputs
Evaluation: AUC, accuracy, precision, recall, F1, κ
Permutation importance, partial dependence, SHAP
30 m suitability maps + ensemble, 5 classes